

Basic Interferometer on Point Sources, the Sun, and the Moon

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Abstract

This report was written for a radio astronomy undergraduate course at UC Berkeley. Radio interferometry is a powerful observational method used in Radio Astronomy for observing the sky in the Fourier domain. In this lab, my partners and I used a local radio interferometer at Berkeley to observe the Sun, the Moon, and point sources. We were also able to approximate the base line of our interferometer and measure the diameter of the Sun and the Moon.

1 Introduction

The purpose of this lab on interferometry was to demonstrate diffraction theory as it applies to a real radio interferometer. Fringes along with their amplitudes and phases were examined along with the “Fourier transform” relationship between the interferometer’s response and luminosity of the sky. Additionally, my partners and I had to learn a lot about time, coordinates, and conversions to progress in this lab. We also obtained horizon-to-horizon data on various astronomical sources ranging from point sources to the Sun and Moon. When performing data analysis, we use linear and nonlinear least-squares fitting methods to filter out accurate point source positions and obtain approximate diameters for the Sun and Moon along with an estimate value of our two-element interferometer’s baseline.

2 Background Theory

2.1 Astronomical Coordinates

There are many different ways to go about keeping track of time and position using coordinate systems. For us to make and record accurate astronomical observations, we must know both where on Earth we are positioned and the current time relative to such location. By knowing both the position and time of observations, the direction from which a telescope is pointed can be accurately deciphered. Latitude and longitude coordinates work well to encode relative positions on the Earth. However because the Earth is not perfectly spherical, detailed surveys using latitude and longitude are not quite as simple as they may appear. On the contrary, accurately keeping track of time has its own issues. On Earth, clocks are tuned to different times according to which time zone and year a person is currently in. However in astronomy, using time zones to keep track of time is not particularly useful. To make counting time easier, astronomers use systems that do away with time zones and synchronize time to the “solar day” at a position of 0° longitude, which also happens to conveniently run through the British Royal Observatory.

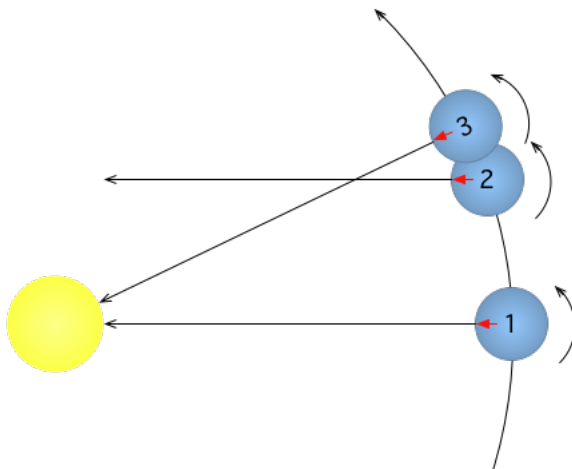


Figure 1: On Earth, because the planet experiences prograde motion, a sidereal day is slightly shorter than a solar day. At position 1, the Sun and a distant star behind are both directly overhead. At position 2, the Earth rotates about 360° , making the distant star behind the sun appear again directly overhead. This shift from position 1 to position 2 is called a sidereal day. However, by the time the Earth rotates back to face the Sun, as seen in position 3, the Sun is again directly overhead, but the distant star behind is not. Since the revolution around the Sun changes the angle from which the Sun is seen relative to the Earth, position 1 to position 3 shows how long it takes the Sun to return to its maximum height at precisely noon everyday. This shift from position 1 to position 3 is called a solar day.

A solar day is the amount of time it takes for the sun to return back to the same position on Earth. Another important thing to keep track of while trying to accurately track time is to count seconds. One might think that counting days is more practical, but in fact, days can be fairly complicated. Since the Earth is moving around the Sun while it is spinning around it's axis, the Earth must actually rotate a little more than 360° , or 2π radians, in order to return to the same position each day. Instead of spinning precisely 360° every 24 hours, or 86400s, it actually takes the Earth about 86164.0908s. This amount of time called a sidereal day and is slightly longer than a solar day (see Figure 1). For every day of the year except January 1, sidereal and solar days are out of sync. When trying to accurately track time, we really should start from January 1. However, the question then becomes what time and year relative to January 1 should one begin counting from?

In astronomy, astronomers typically count time in solar days starting from noon, January 1, 4713 BC, also known as the Julian date. On the other hand, computers count time in seconds from midnight at the start of January 1, 1970. This is called "Unix Time" and is pretty close to Coordinated Universal Time (UTC) which is derived directly from atomic clocks. Today, there have been a few revisions to UTC with several leap seconds added in here and there. Additionally, because the Earth's rotation is slowly slowing down, and keeping time can become extremely complicated. By using Network Timing Protocol (NTP) and/or the Global Positioning System (GPS), computers and cell phones routinely synchronize their clocks to atomic standards with milliseconds of precision. Because UTC is widely available to most people today, this makes it relatively easy for us Astronomers to accurately track time.

2.2 Coordinate Conversion

2.2.1 Longitude and Latitude

Longitude and latitude coordinates are considered geographic coordinates (see Figure 2). A geographic coordinate system is a coordinate system that enables any location on the Earth to be specified by a set of numbers or letters. The latitude of a point on the Earth's surface is the angle between the equatorial plane and the straight line that passes through that point normal to the surface. Lines joining points of the same latitude form circles on the surface of the Earth known as parallels because they are parallel to the equator and to each other. While the $+90^\circ$ parallel represents the north pole and -90° parallel represents the south pole, the 0° parallel of latitude is at the equator which divides the globe into Northern and Southern Hemispheres.

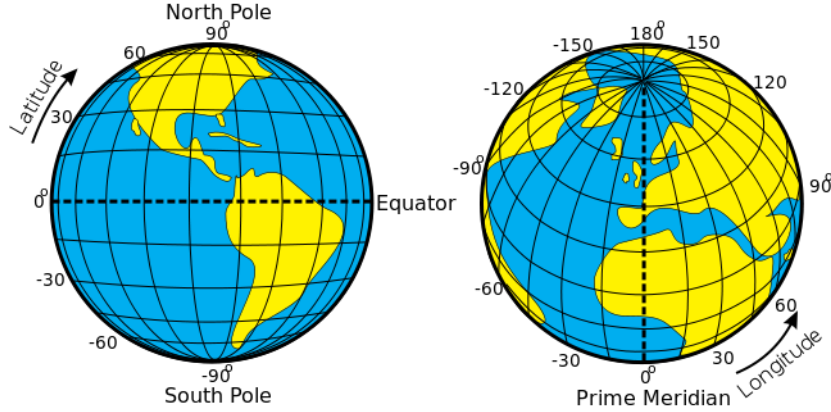


Figure 2: On Earth, latitude coordinates are measured as angles from the Equator to $+90^\circ$ (North Pole) to -90° (South Pole). Longitude coordinates are measured from the Prime Meridian from 0° to $+180^\circ$ due right and from 0 to -180° due left.

The longitude of a point on the Earth's surface is the angle east or west from the reference meridian to another meridian that passes through that point. The Prime Meridian is the line chosen as the international zero-longitude reference line and was set with the intention of passing through the British Royal Observatory in Greenwich, UK. Places to the east of the Prime Meridian are considered to be in the eastern hemisphere while places to the west are in the western hemisphere. Longitude also ranges from about -180° to $+180^\circ$ relative to the Prime Meridian.

Additionally, converting from rectangular latitude and longitude coordinates to spherical coordinates can be done using the vector matrix shown below in Equation 1.

$$\vec{x} = \begin{bmatrix} \cos(lat) \cos(long) \\ \cos(lat) \sin(long) \\ \sin(lat) \end{bmatrix} \quad (1)$$

2.2.2 Right Ascension and Declination (α, σ)

In spherical coordinates, the position of stars are often expressed in right ascension and declination without use of a coordinate for distance. Because celestial objects are extremely far away, astronomers often have little to no information about their exact distances and therefore use only

the object's direction. Additionally, the direction of fairly distant objects is the same for all observers and it is convenient to specify this direction with the same coordinates for every object. On the contrary, through the horizontal coordinate system, a star's position differs from observer to observer based on their positions on the Earth's surface and continuously changes with the Earth's rotation.

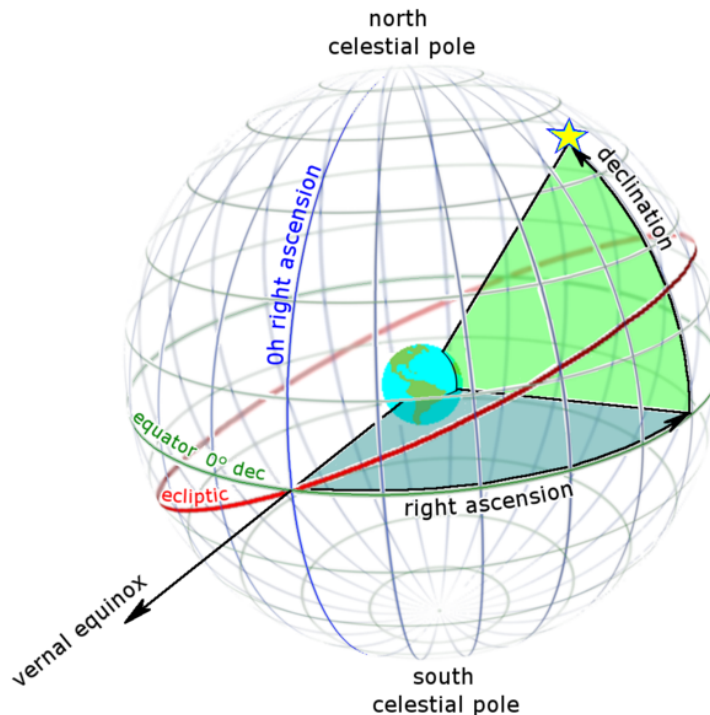


Figure 3: The figure above shows right ascension and declination as seen on the inside of the celestial sphere. The primary direction of the system is the vernal equinox or where the ascending node of the ecliptic in red meets the celestial equator in blue. Right ascension is measured eastward along the celestial equator from the primary direction while declination is measured due north or south from the equator.

The equatorial coordinate system is a widely used celestial coordinate system used to specify the positions of celestial objects. Two coordinates typically used in the equatorial coordinate system are the right ascension and declination (see Figure 3). Right ascension (symbol α , abbreviated RA) measures the angular distance of an object eastward along the celestial equator from the vernal equinox to the hour circle passing through the object. The vernal equinox point is one of the two points where the ecliptic intersects with the celestial equator. In comparison to longitude, right ascension is similarly but is measured in sidereal hours, minutes and seconds instead of degrees. This results from measuring right ascensions by timing the passage of objects across the meridian as the Earth rotates. In total, there are $(360^\circ / 24\text{h}) = 15^\circ$ in one hour of right ascension and 24h of right ascension around the entire celestial equator.

The celestial equator is the projection of the Earth's equator onto the celestial sphere. Declination (symbol δ , abbreviated Dec) measures the angular distance of an object perpendicular to the celestial equator. Declination ranges from $+180^\circ$ northward to -180° to the south. The north celestial pole, for instance, has a declination of $+90^\circ$. The origin of the coordinate of declination is the celestial equator itself. Declination can also be thought of as analogous to latitude. When

the two coordinates are used together, right ascension and declination are typically abbreviated as (RA, Dec) or (α, σ) . Because RA and Dec are analogous to longitude and latitude, Equation 1 on page 3 can also be used to convert RA and Dec to spherical coordinates.

2.2.3 Hour Angle and Declination (ha, σ)

In the equatorial coordinate system, right ascension can also be translated to hour angle (abbreviated HA or LHA). Hour angle positioning is a “left-handed” system which measures the angular distance of an object westward along the celestial equator from the observer’s meridian to the hour circle passing through the object (See Figure 4).

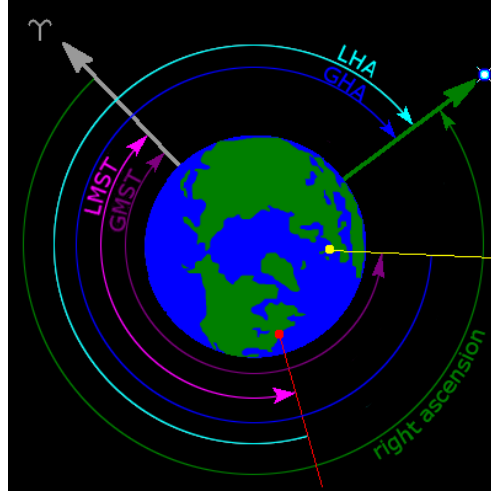


Figure 4: The figure above shows right ascension measured eastward along the celestial equator from the primary direction along with the hour angle abbreviated by LHA. The hour angle coordinate is the right ascension summed with the local sidereal time.

Unlike right ascension, hour angle increases with the Earth’s rotation. Hour angle is used to measure the time from when an object crosses the meridian. For instance, a star on the observer’s celestial meridian is said to have a zero hour angle. About one sidereal hour later, the Earth’s rotation will carry the star to the west of the meridian and its hour angle becomes +1h. Right ascension may also be converted into hour angle as an intermediate step to converting to azimuth and altitude by multiplying the spherical coordinates with the rotational matrix shown below in Equation 2.

$$R_{(\alpha, \delta) \rightarrow (ha, \sigma)} = \begin{bmatrix} \cos(LST) & \sin(LST) & 0 \\ -\sin(LST) & \cos(LST) & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (2)$$

2.2.4 Azimuth and Altitude (az, alt)

The horizontal coordinate system is a celestial coordinate system that utilizes an observer’s local horizon as it’s fundamental plane. This coordinate system divides the sky into an upper hemisphere and a lower hemisphere. In the upper hemisphere, objects and stars are visible or within sight while in the lower hemisphere, the our view of the sky is blocked by the Earth. The “great circle”

separating hemispheres is called celestial horizon or rational horizon and the pole of the upper hemisphere is called the zenith.

Altitude (abbreviated Alt) is sometimes referred to as elevation or the angle between the object and the observer’s local horizon. For visible objects in the sky, altitude is an angle between 0° to 90° . Azimuth (abbreviated Az) is the angle of the object about the horizon and is usually measured from the north while increasing towards the east. Because the horizontal coordinate system is fixed to the Earth and not the stars, the altitude and azimuth of an object changes with time as the object appears to “drift” across the sky. Furthermore, because the horizontal system is defined by the observer’s position or local horizon, different locations on Earth viewing the same object at the same time will have different values of altitude and azimuth.

By multiplying spherical hour angle and declination coordinates by the rotational matrix shown below in Equation 3, we can convert (ha, σ) coordinates to (az, alt) coordinates. Since hour angle and declination are “Earth-based” coordinates, this conversion will only depend on the observer’s terrestrial latitude ϕ . Additionally, we can also convert directly from right ascension and declination to azimuth and altitude by multiplying our coordinates by both rotational matrices.

$$R_{(ha,\delta)\rightarrow(az,alt)} = \begin{bmatrix} -\sin(\phi) & 0 & \cos(\phi) \\ 0 & -1 & 0 \\ \cos(\phi) & 0 & \sin(\phi) \end{bmatrix} \quad (3)$$

2.3 Interferometry

Interferometry is the practice of using two or more radio telescopes in a synchronized array to observe astronomical sources and make data measurements. A basic interferometer is composed of the array of radio telescopes along with the electronics used to measure, record, and detect signals. For this lab, my partners and I used a narrow band two-element interferometer located on the roof of Kroeber Hall at UC Berkeley, separated along an approximately East-West baseline. Additionally, the interferometer operates at about 10.67 GHz, the baseline separation between the two dishes is approximately $B = 10\text{-m}$, and the observe radiation is at a wavelength of $\lambda = 2.5\text{ cm}$.

2.3.1 The Two-Element Interferometer

A two-element interferometer (see Figure 5 on the next page) consists of two telescopes separated by a vector $\vec{b}$ known as the baseline. In this system, both antennas simultaneously receive electromagnetic radiation from a specified astronomical source located above in the sky in the direction of the unit vector $\hat{s}$. Since astronomical sources are extremely far away, radiation is transmitted to the antennas in the form of “plane waves.” A plane wave is a constant-frequency waveform in which the wavefronts are infinite parallel planes of constant peak-to-peak amplitude normal to the phase velocity vector.

$$\vec{b} \cdot \hat{s} = |\vec{b}| \cos \theta, \quad \tau = \frac{\vec{b}_{ij} \cdot \hat{s}}{c} \quad (4)$$

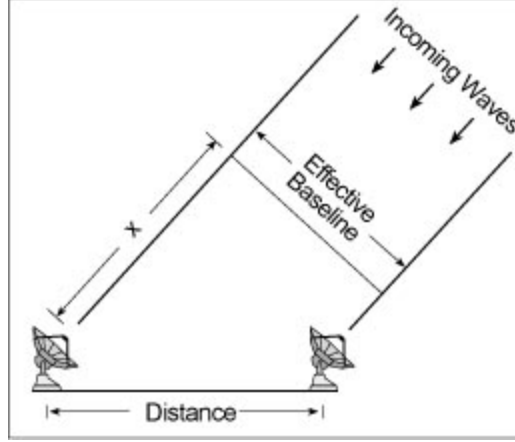


Figure 5: When an incoming radio wave approaches the Earth at an angle, the radio wave will arrive at the two separated telescopes at slightly different times. In this picture, the delay in arrival time, also known as the geometric delay, is the distance X divided by the velocity of light. Also, the angular resolution of the interferometer is equal to the wavelength divided by the effective baseline of the telescopes. When the object being observed is directly overhead, the effective baseline is approximately equal to the distance between the two telescopes.

As the plane waves from astronomical sources reach the radio dishes, one antenna will end up receiving the signal before the other. The time that passes between the receiving of the signals is called the geometric delay and is denoted with the variable τ . In order to accurately find the value of τ , we must know the “extra distance” that the plane waves have to travel before reaching the second telescope. This distance is approximately equal to the baseline vector dotted with the directional unit vector. Additionally, since the direction of the source is a unit vector, this distance is approximately the projection of the baseline vector onto the unit vector (see Equation 4 on the preceding page) with θ being the angle between the baseline and direction vectors. If the baseline projection of the interferometer is known, then the geometric delay τ can be calculated by dividing this distance by the velocity of the plane waves (the speed of light, c). When this delay is an exact multiple of the wavelength, then the waves detected by the two telescopes will be in phase and will add together when combined. If they are not, then they will be out of phase and interfere with one another.

If there is another source in the sky at a different direction from the first, the second source will also cause a geometric delay between the two antennas. However, this would be different from the geometric delay of the first source because it would have different directional unit vectors $\hat{s}$. Because each antenna is receiving a signal from each source, what is detected is actually the total signal of the two. In order to separate the signals from each source at each radio dish, we must correlate the total signals with each other.

2.3.2 Correlation

Correlation is very similar to convolution theorem. However, there is a slight difference. When correlating two signals, the Fourier transform of the second function is conjugated before being multiplied and integrated over. Additionally, by correlating a flat-spectrum signal with a time-shifted version of itself, this will yield a measure of the power of the signal at the delay corresponding to the time shift. For a two-element interferometer, the total signal detected by each antenna can be represented as shown in Equation 5 on the next page, where e_1 and e_2 are the signals given by

each source at times t and the difference between t and either τ_1 or τ_2 .

$$E(t) = e_1(t) + e_2(t), \quad E_j(t) = e_1(t - \tau_1) + e_2(t - \tau_2) \quad (5)$$

$$(f \star g)(\tau) = \int f(t)g^*(t - \tau)dt \quad (6)$$

By using $E_i(t)$ and $E_j(t)$ as f and g in the correlation equation (see Equation 6) respectively, the correlation equation then becomes:

$$(f \star g)(\tau) = \int e_1(t)e_1(t - \tau_1 - \tau) + e_2(t)e_2(t - \tau_2 - \tau)dt \quad (7)$$

In order to extract meaningful signal information out of the integral shown above in Equation 7, the two functions must both not average to zero. This is only possible when τ is equal either to $-\tau_1$ or $-\tau_2$. In the case of $\tau = -\tau_1$, the correlation integral of the two sources becomes:

$$(f \star g)(\tau) = \int e_1(t)e_1(t - \tau_1 - (-\tau_1))dt = \int e_1(t)e_1(t)dt = \langle e_1^2 \rangle \quad (8)$$

In this expression, the e_2 term averages to zero because $\tau_2 - (-\tau_1)$ does not cancel. The final result is the average power received by the antennas for either the first or second source depending on the geometric delay τ . So for a set of different geometric delays, there is a set of power values that rises above signals that cannot be correlated. The set of power values are essentially noise.

2.3.3 Visibility Equation

A single frequency ν with corresponding wavelength λ has a number of wavelengths according to the relationship to the baseline $\tau_{ij}\nu = \frac{\vec{b} \cdot \hat{s}}{\lambda}$. By knowing the number of wavelengths between the two antennas, we can say that for a signal of a particular frequency coming from a source in direction $\hat{s}$, the complex phase difference between that signal measured at antenna i and antenna j will be $e^{i\vartheta}$, where ϑ is the angle covered by the wave as it propagates from i to j . Since ϑ is 2π times the number of wavelengths, we can the phase difference $\Delta\phi$ (see Equation 9).

$$\Delta\phi = e^{-2\pi i\tau_{ij}\nu} = e^{-2\pi i\frac{\vec{b} \cdot \hat{s}}{\lambda}} \quad (9)$$

The phase difference $\Delta\phi$ is as shown in the equation above frequency dependent. At particular frequencies, $\Delta\phi$ also varies with position across the sky. The pattern traced by the complex phase difference on the sky is called the “fringe pattern” of the interferometer. To extrapolate from a single baseline in a single direction a picture of how an entire array might respond to the entire sky that falls within the primary beam of the correlated telescopes, we must define coordinates representing the length of a baseline in units of wavelength (see Equation 10). Additionally, we must split the source direction vector $\hat{s}$ into its components.

$$\frac{\vec{b}}{\lambda} \equiv (u, v, w), \quad \hat{s} \equiv (l, m, \sqrt{1 - l^2 - m^2}) \quad (10)$$

In Equation 10, u represents the east-west component of the baseline, v the north-south component, w the vertical (up-down) component, l the east-west direction on the sky, and m the north-south direction. The $\sqrt{1 - l^2 - m^2}$ component of $\hat{s}$ comes from the fact that we restrict our direction to be a unit vector. Using these components, the response of a baseline, or “visibility” V ,

as a function of the u, v, w separation of the antennas, integrated over all source intensity I on the sky as a function of l and m can be written as follows:

$$V(u, v) = \int \int A(l, m) \cdot I(l, m) \cdot e^{-2\pi i(ul+vm+w\sqrt{1-l^2-m^2})} dl \, dm \quad (11)$$

Equation 11 is the complete form of the “visibility equation” and is also sometimes referred to as the “measurement equation” of an interferometer. Variable $A(l, m)$ is the response of the primary beam of the antennas as a function of it’s direction on the sky. In general, $A(l, m)$ and $I(l, m)$ are grouped together because the sky is always perceived through the filter of the primary beam. The product between the two is sometimes referred to as the “perceived intensity” of the beam.

Equation 11 can be easier understood if we make a simplifying assumption known as the “flat-sky” approximation. The “flat-sky” approximation is either that $w = 0$ or that the primary beam $A(l, m)$ is sufficiently small that $l, m \ll 1$, making the component $\sqrt{1-l^2-m^2} \approx 1$. For either approximation, the response of a baseline does not have to account for the fact that the sky is a “curved surface” of a sphere. With this assumption, the $e^{-2\pi i w \sqrt{1-l^2-m^2}}$ term is no longer a function of l and m and can therefore be removed from the integral (see Equation 12).

$$V(u, v) = e^{-2\pi i w} \int \int A(l, m) \cdot I(l, m) \cdot e^{-2\pi i(ul+vm)} dl \, dm \quad (12)$$

This version of the visibility equation shown above states that when the telescopes are phased to a “phase center” through a choice of a corresponding component $e^{-2\pi i w}$ with w being the baseline component along the direction toward the phase center in units of wavelength, the visibility $V(u, v)$ is becomes just the Fourier Transform of the perceived sky (see Figure 6).

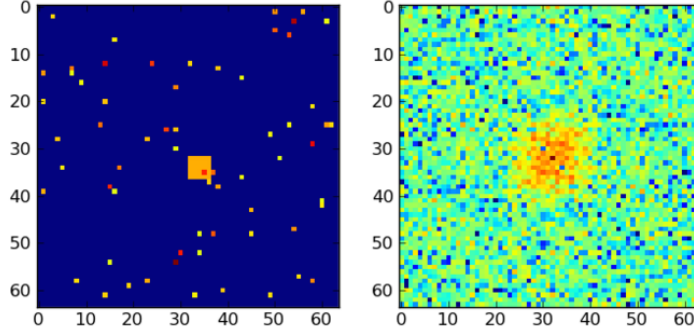


Figure 6: The true image of the sky is shown on left and the true UV plane is shown on the right.

To better understand the fringe-pattern of a baseline on the sky, we can equivalently think of taking an image of the sky in l, m coordinates and then Fourier transform to create a “uv-plane” with coordinates being “inverse angles.” Because an inverse angle is the same thing as a wavelength, the uv-plane will have coordinates of u, v . Furthermore, for each pair of antennas, there will be two samples; one at u, v and another at $-u, -v$.

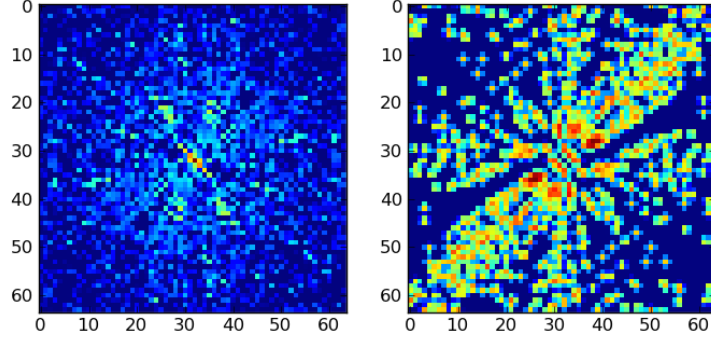


Figure 7: The array sampling pattern in the uv -plane is shown on the right and the associated synthesized beam pattern or “dirty beam” on the left.

Since the sky is real-valued, these two Fourier components are related by a complex conjugate. This means that if we measure $V(u,v)$ at u,v , we will also have to measure $V^*(u,v)$ at $-u,-v$. The sampling of the uv -plane is done by multiplying the true uv -plane with the sampling pattern computed. This result is what we would get if we took visibilities measured from an interferometer and then placed each visibility $V(u,v)$ at the corresponding u,v (along with $-u,-v$) coordinates of a matrix. Moreover, if you then take the inverse Fourier Transform of this sampled uv -plane, you get a dirty image (see Figure 8).

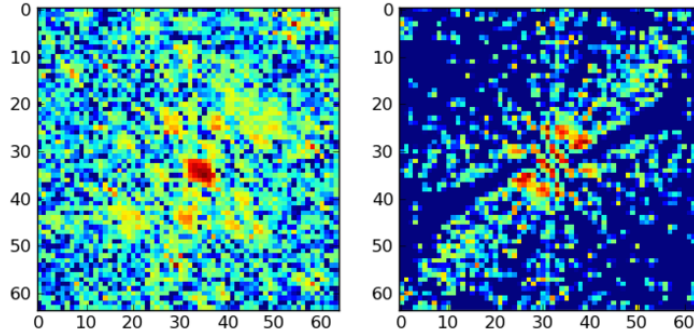


Figure 8: The array sampling pattern in the uv -plane is shown on the right while the associated synthesized beam pattern or “dirty beam” is on the left.

Because the image shown in Figure 8 is fairly degraded from the original, this is usually called a “dirty image”. Information is thus lost while sampling the uv -plane. If we multiply the true uv -plane by our sampling function, this is equivalent to convolving the true sky with the Fourier Transform of our sampling function, also known as the “dirty beam”. The dirty beam is what is convolved with the true sky to yield our dirty image. Additionally, it is possible to recover something resembling the true sky by deconvolving the dirty image with the dirty beam (see Figure 9 on the next page).

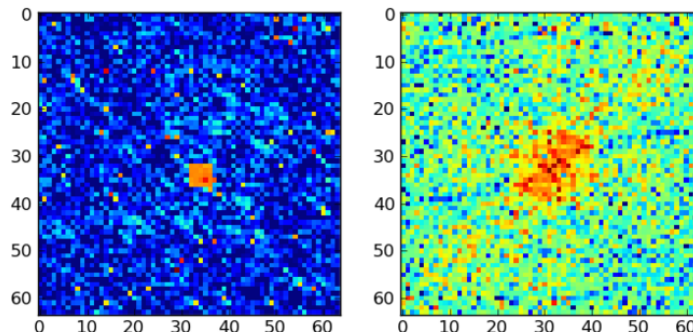


Figure 9: This figure shows the result of a “cleaned” dirty image. Note that the point sources are recovered but the extended square in the middle is slightly worse as a result of using a “point-source model” in the deconvolution process.

Deconvolution attempts to compensate for the information that is lost by sampling part of the uv-plane by inserting prior information about the sky. This information can be along the lines of just knowing that the sky is a set of point-sources to knowing that we want the smoothest sky that fits our data to the level of noise. Nevertheless, deconvolution is not a well-posed issue until you decide precisely what prior information you want to use for the sky.

3 Methods and Observations

3.1 Using the Interferometer

In this lab, the interferometer used in this lab has a relatively short baseline and the fringe spacing is larger than the size of all sources except for the Sun and Moon. Because of this, all of these sources looked like “point sources” from which all the radiation appears to come from a single point like a star in optical astronomy. The output of our interferometer is a sinusoidal-like signal called the “fringe”. All information for the data collected using the interferometer is stored in the frequency, amplitude, and phase of the fringe. Additionally, if we initially know the baseline, the fringe properties become indicators of the point-source’s declination.

The interferometer we used in this lab is also called a multiplying interferometer. A multiplying interferometer takes the signals from the two telescopes and multiplies them together using a mixer. We repetitively measured and recorded the time average over a suitable interval. Using the cross product between the signals is important because its average value is zero unless there is a source. Thus, any detected signal has to come from the sky instead of from the instrument itself. When we observe a source, the fringe amplitude has no zero offset and the amplitude is therefore directly proportional to the source’s brightness or flux.

3.2 Measuring Accurate Declinations

Since our interferometer has a baseline of about 10-m, we can recover a fringe spacing of about $\frac{\lambda}{B} \sim 10'$. From a horizon-to-horizon measurement, we were also able to measure the declination approximately 100 times more accurately. Furthermore, we can also achieve partial angular resolution of the Sun and Moon along with their diameters to a fraction of a percent.

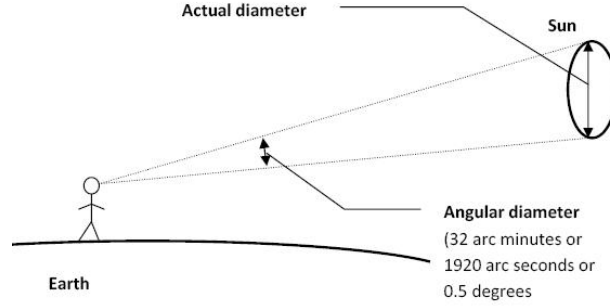


Figure 10: The figure above illustrates the concept of “angular diameter” in units of arc minutes, arc seconds, and degrees.

3.3 Object Tracking in Python

To track astronomical sources with the interferometer using the programming language Python, my lab partners and I used the “PyEphem” and “Radiolab” Python libraries to position our telescope and monitor time. By using UTC, we were able to figure out what sidereal time it is in England at a longitude of 0° in hours, minutes, and seconds. Since right ascension is measured in units of time, this told us what part of the sky is overhead at the prime meridian. From our current longitude at Berkeley (37.8732° N, -122.2573°), we figured out the time offset we had add to get the current local sidereal time. The LST tells us what part of the sky is directly overhead at our location. By subtracting from the RA the LST of our astronomical sources, we were able to get the hour angle as it appears from our location. Finally, from using the latitude, HA, and the declination of our sources, we were able to figure out precisely how and where to point the telescope.

3.4 Point Sources

While choosing point sources, my partners and I had to be careful. Since the fringe frequency is dependent on $\cos \delta$, this means that we did not measure δ directly, but rather $\cos \delta$. Southern sources presented a minor problem for horizon-to-horizon tracking because of the presence of the Campanile at UC Berkeley. To compensate for this, we had to delete some of our data. Additionally, our telescope looking for Northern sources may be inaccurate, leading to loss of the signal and noise. Our observation frequency was also in the TV satellite band. Because geostationary TV satellites resides in the southern skies, they can generate strong signals that could enter the telescope’s dish sidelines and produce fringes.

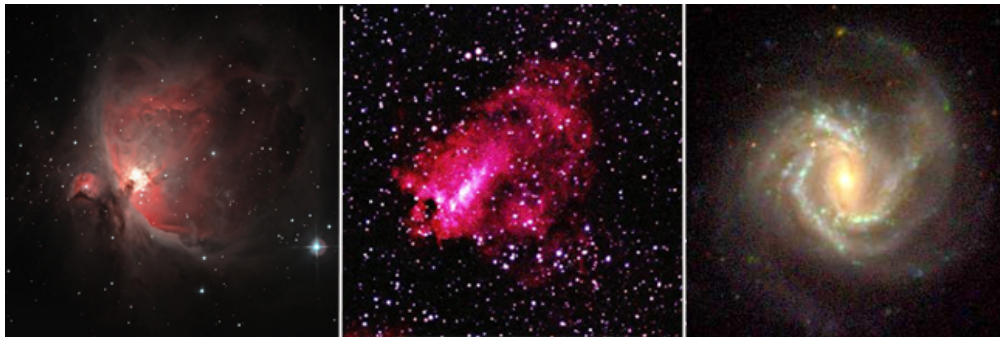


Figure 11: Pictures of Orion Nebula, M17 Nebula, and Virgo A respectively.

The three sources that my partners and I decided to observe were the Orion Nebula, M17 Nebula, and Virgo A. From our observations, my partners and I measured the approximate baseline of our interferometer using fixed declination values for each source. This was done by using the method of “least-squares fitting” in Python on the fringe phase to the hour angle on fringe amplitude Equation 13 shown below.

$$F(h_s) = A \cos \left[\left(2\pi \left(\frac{B_y}{\lambda} \cos(\delta) \right) \sin(h_s) \right) \right] - B \sin \left[\left(2\pi \left(\frac{B_y}{\lambda} \cos(\delta) \right) \sin(h_s) \right) \right] \quad (13)$$

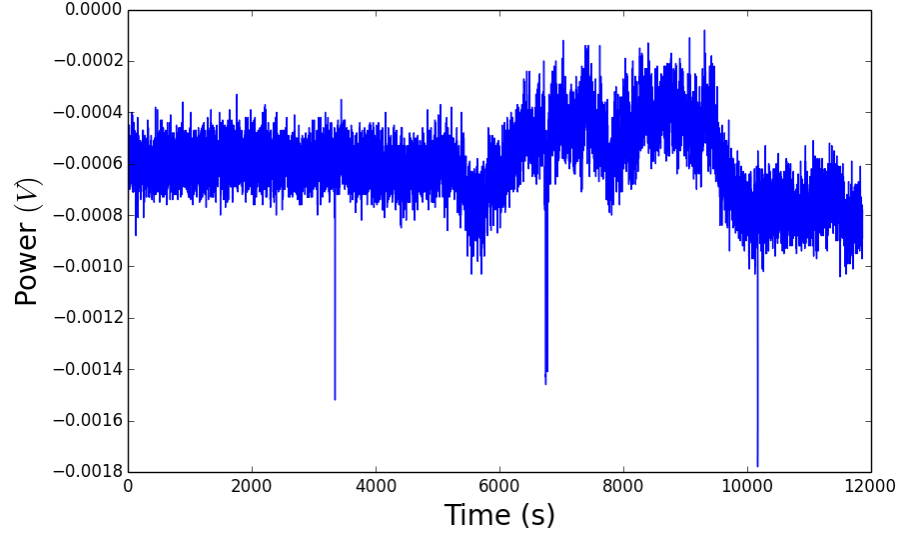


Figure 12: Untreated data from the Orion Nebula, collected from March 21, 2014 10:53PM - March 22, 2014 2:20AM.

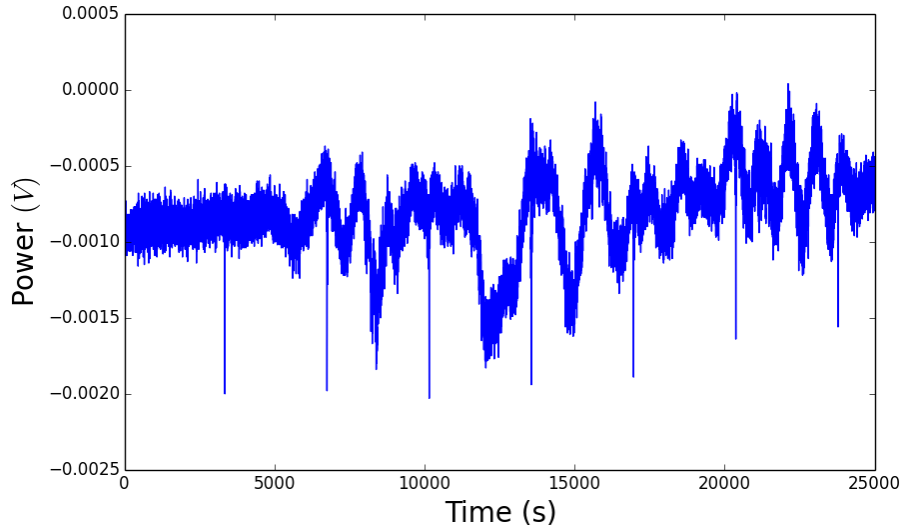


Figure 13: Untreated data from the M17 Nebula, collected from April 5, 2014 10:07AM - 5:04PM.

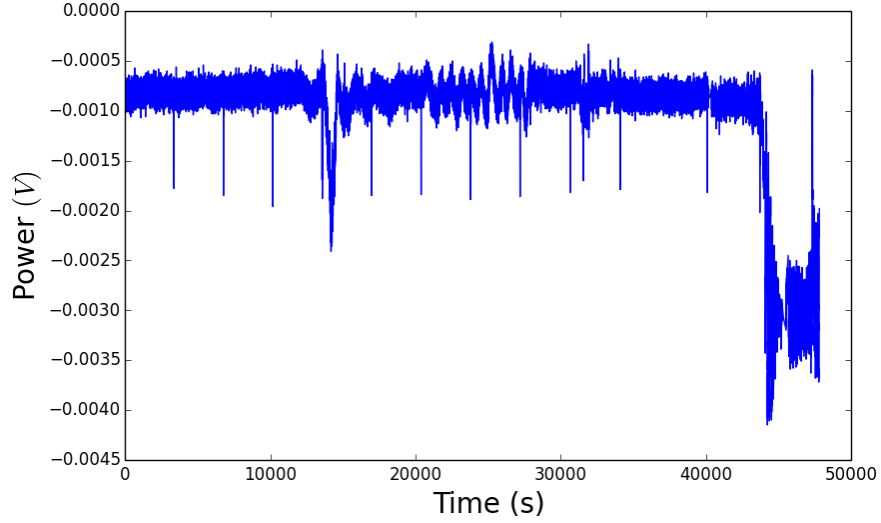


Figure 14: Untreated data from Virgo A, collected from April 3, 2014 3:03AM - 4:20PM.

3.5 The Sun and Moon

The Sun and Moon are special cases of astronomical sources for two reasons. First of all, the position of the Sun and Moon change from day to day. For the Moon, even a change of just an hour is extremely significant. Secondly, the Sun and Moon are both extended sources, not point sources. This causes their fringe amplitudes and phases to change with time depending on their brightness distribution and can make the determination of accurate positions somewhat complex. Nevertheless, my partners and I were able to get the Sun and Moon positions directly from the Python library “PyEphem”.

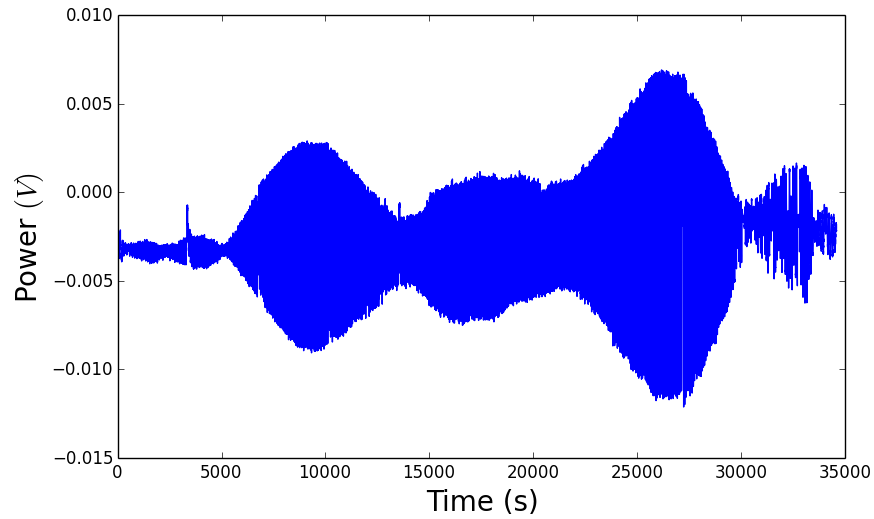


Figure 15: Untreated data from the Sun, collected from April 1, 2014 3:25PM - April 2, 2014 1:02AM.

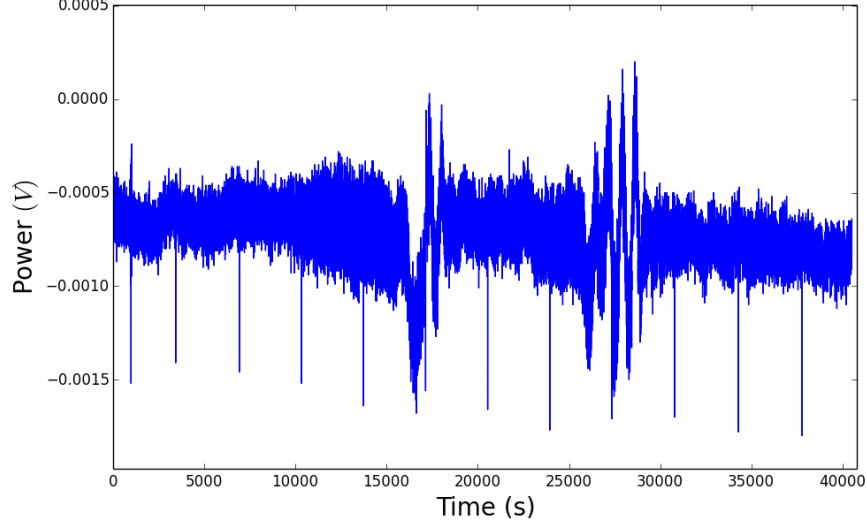


Figure 16: Untreated data from the Moon, collected from April 4, 2014 6:54PM - April 5, 2014 6:09AM.

Since the Earth's Moon is relatively nearby, we also had to correct for our location on the Earth's surface. The term "parallax" refers to the semi-angle of inclination between two sight-lines to the star as observed when the Earth is on opposite sides of the Sun in its orbit. The parallax effect for the Moon is far greater than that of the Sun. Unless if this is corrected for, the Moon will probably lie outside the telescope's main beam. Fortunately, PyEphem also automatically takes care of this along with other complex issues like the "time-of-flight" of photons. Regardless of these difficulties, the Sun and Moon are still good sources because they are really bright and included them in our observations because they produce huge signals which is ideal for testing our observations and data analysis. Furthermore, the Sun is so bright that we got a huge signal to noise ratio. This allowed us to estimate an accurate declination.

From our observations, my partners and I measured the approximate phase offset ϕ of our interferometer using the average $\frac{B_y}{\lambda}$ calculated from the point source data each source. This was done by using the method of "least-squares fitting" in Python on the fringe phase to the hour angle on fringe amplitude Equation 14 shown below.

$$R(h_s) = \cos\left(\frac{2\pi B_y}{\lambda} \cos(\delta) \sin(h_s) + \phi\right) \quad (14)$$

3.6 Additional Details

Since the two-element interferometer telescope has different distances for each antenna from the source, it's convenient to think of the different distances in terms of relative path delay in units of time for the two telescopes. As stated earlier, this is called the geometrical path delay τ_g . However, the signals have to travel through a lot of electronics before they are finally multiplied. Additionally, these two paths are not necessarily of equal length. This makes there an additional relative delay from the difference in cable length τ_c . The total relative delay is the sum of the two (see Equation 15). While τ_g is a function of time, τ_c is independent of time.

$$\tau_{total} = \tau_g(h_s) + \tau_c, \quad \tau_g(h_s) = \left[\frac{B_y}{c} \cos \delta\right] \sin h_s \quad (15)$$

The local fringe frequency is the frequency of a certain point on the sky. To see this, we can expand the hour angle term $\sin h_s$ into a Taylor series centered about the current hour angle of the source h_s (see Equation 16). The local fringe frequency is contained in the second term of Equation 16 because for a small region around h_s in Equation 13 on page 13, $F(h)$ varies as $f_f \Delta h$. The local fringe frequency f_f is shown in Figure 17.

$$\sin h = \sin h_s + \Delta h \left. \frac{d \sin h}{dh} \right|_{h_s} = \sin h_s + \Delta h \cos h_s \quad (16)$$

$$f_f = \frac{C}{2\pi} \cos(h_s) = \left[\frac{B_y}{\lambda} \cos \delta \right] \cos h_s \quad (17)$$

Equation 17 represents the local fringe frequency in cycles per radian on the sky. To convert to cycles per hour, we can multiply $\frac{dh_s}{dy} = \frac{2\pi}{24}$ where the number 24 represents the number of hours in a day. For instance, at the meridian, assuming $B_y \approx 10m$ and $\lambda = 2.5cm$, the fringe frequency is approximately $0.029 \cos \delta$ cycles per second while the period is $\frac{1}{f_f} = \frac{35}{\cos \delta} = 35s$.

3.6.1 Least Square Fitting

A least squares method can be used on the filtered data from the point sources and from the Sun and Moon using the fringe amplitude (Equation 13 on page 13 and 14 on the previous page). For instance, for the point sources, this is done by minimizing the square of the residuals (χ^2) for particular values of τ_c and B_y . Using the following procedure shown below, my lab partners and I were able to fit for both τ_c and B_y .

(1) Using python, we inserted Equation 13 on page 13 with guessed constant coefficients for the τ_c dependent factors.

(2) Next, we iterate over a range of possible baseline values B_y , and apply a least squares fit to each value. This provided us with a χ^2 and values of A and B.

(3) By looking at the specific value of B_y corresponding to the minimum χ^2 value, this gives us the “least squares fit” for B_y , A, and B.

3.6.2 Measuring The Diameter Of A Circular Source

In this lab, goal was to measure and compare the diameters of the Sun and Moon. Assuming that the sources are uniformly-bright disks of radius R, this makes the intensity distribution approximately equal to Equation 18.

$$I(\Delta h) = \frac{(R^2 - \Delta h^2)^{1/2}}{R} \quad (18)$$

Additionally, to obtain the theoretical modulating function MF_{theory} , we can use the following expression:

$$MF_{theory} = \frac{1}{R} \int_{-R}^R (R^2 - \Delta h^2)^{1/2} \cos(2\pi f_f \Delta h) d\Delta h \quad (19)$$

If we want to calculate MF_{theory} analytically, this can be done by substituting $\Delta(R \cos \theta)$ for Δh . The result is a Bessel function. To do this, we can split $I(\Delta h)$ into $2N+1$ tiny slices. Since the total number is odd, this makes the slices symmetric about $\Delta h = 0$. Each of these slices has width $\delta h = \frac{R}{N}$ along with $\Delta h_n = n\delta h$, where n ranges from $-N$ to $+N$. After being rewritten, the integral becomes the sum shown below.

$$MF_{theory} \approx \delta h \sum_{n=-N}^{n=+N} \left[1 - \left(\frac{n}{N} \right)^2 \right]^{1/2} \cos\left(\frac{2\pi f_f R n}{N}\right) \quad (20)$$

4 Data Analysis

4.1 Python Coding to Clean Data

To treat the data my partners and I gathered, I made a Python script to eliminate bad data points for all my data, normalize the plot to the origin, fit an envelope for the Sun and Moon data, plot the power spectrum, and perform a “Least Square Mean” curve fit to return the baseline B_y for point sources, and use polynomial curve fitting for the a parabolic portion of the Sun and Moon to derive a Bessel function, and calculate the angular diameter of the sources. Additionally, my script was also built to make plots for just about any datasets taken from the interferometer given a specified file path.

To normalize datasets to the x-axis, I using a method of data smoothing called “Boxcar Averaging”. This was done by averaging 100 points above and below each datapoint to create an array of average data. Initially, I used the Boxcar Average to find data points to cut out of the raw data before entering the Boxcar average. Then, by subtracting out the average data plot from the original graph, the data becomes centered about roughly “y = 0”. This “cleaning” my data data calculations and analysis possible.

The next thing I did was fit a curve to the envelope of the Sun and Moon’s normalized data. To find the envelop, I cut out from the data points points whose derivative was greater than a number relatively close to 0. This allowed me to only plot points along the peaks in the signal. Then, by taking the absolute value of the truncated data and using another Boxcar average on the data points to smooth the data, I was able to get a nice envelope for the data. For the Moon, however, because the data points had a lot more spikes, I had to use a scaling factor of about 2.7 to scale my envelop to the correct amplitude.

After finding the envelop curve for my Sun and Moon data, I was then able to fit a parabolic polynomial curve to a “parabola like” region of my data. By using modulation function shown in Equation 20 along with the constants taken from the fitted polynomials, I was then able to derive a Bessel function to approximate the diameter of the Sun and the Moon.

4.2 Sun and Moon Data

4.2.1 The Sun

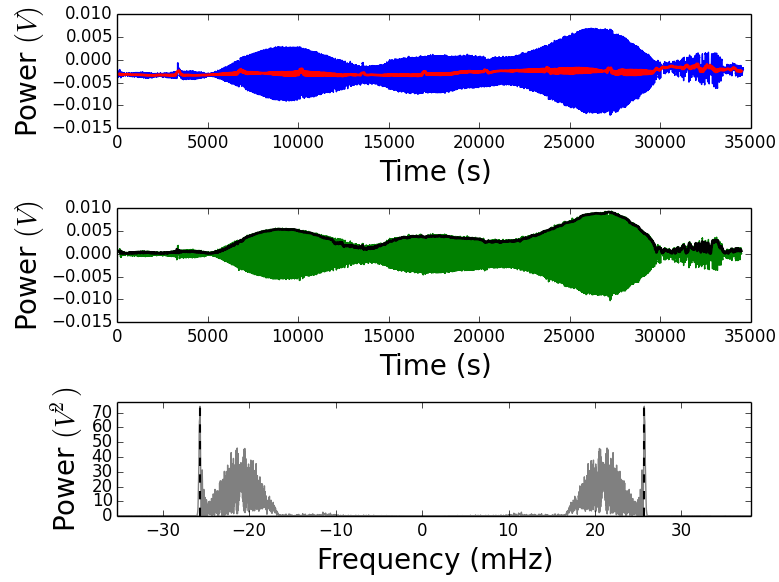


Figure 17: Treated data with envelope and power spectrum from the Sun. Power leakage occurred at about 25.71mHz.

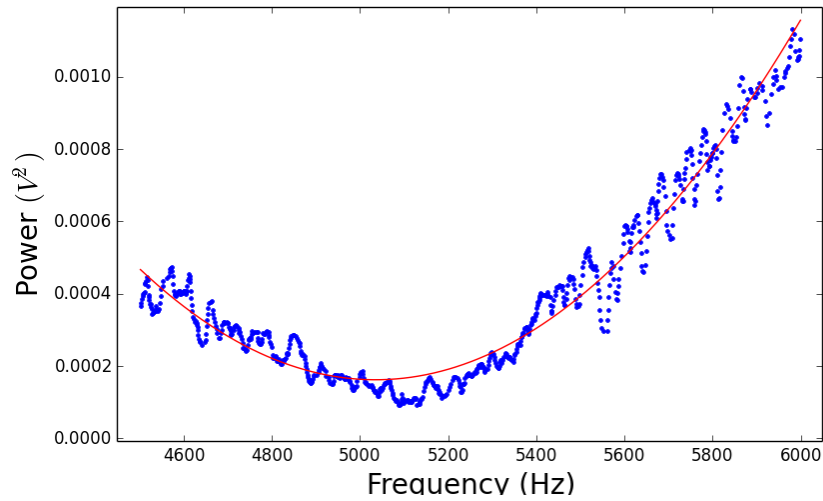


Figure 18: This figure shows a polynomial curve fit for the Sun data.

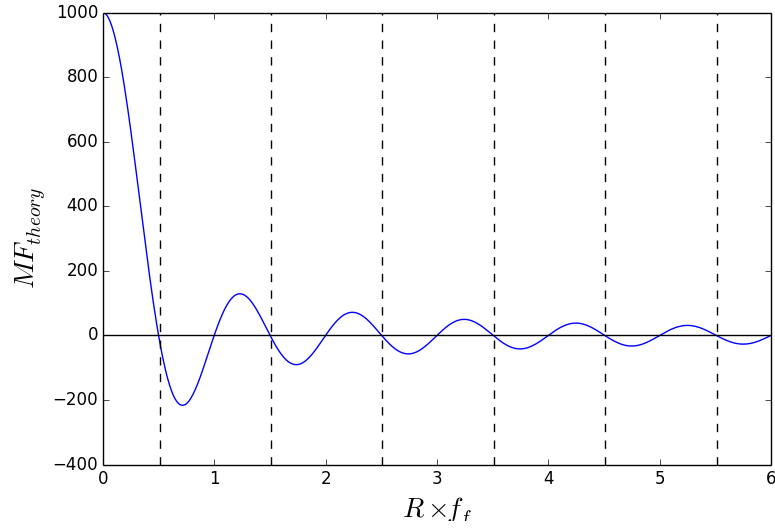


Figure 19: This figure shows the Bessel function for our Sun data comparing MF_{theory} from Equation 20 on page 17 to $R \times f_f$ in order to calculate the radius. Values used for $R \times f_f$ are when the function above crosses the x-axis.

4.2.2 The Moon

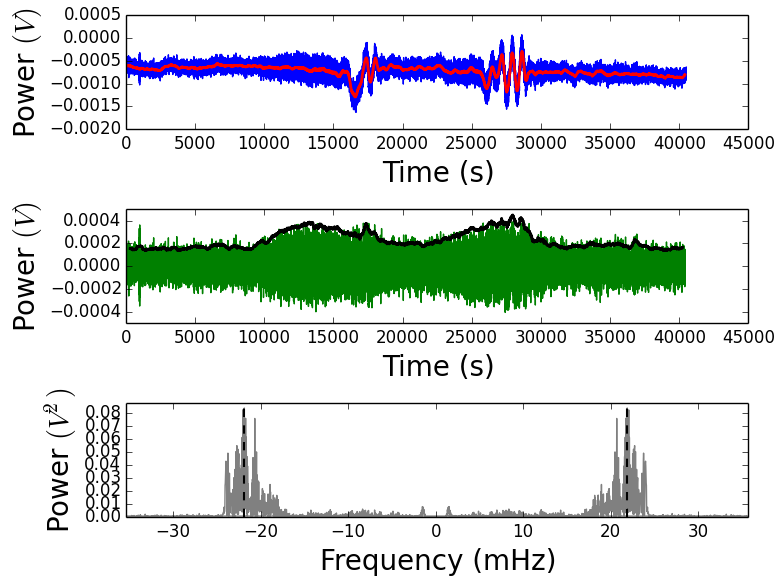


Figure 20: Treated data with envelope and power spectrum from the Moon. Power leakage occurred at about 21.86.71mHz.

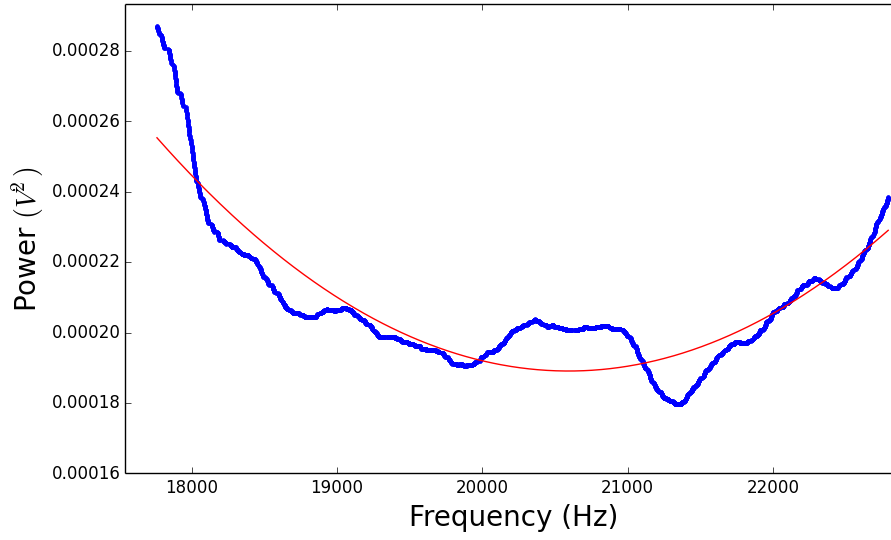


Figure 21: This figure shows a polynomial curve fit for the Moon data.

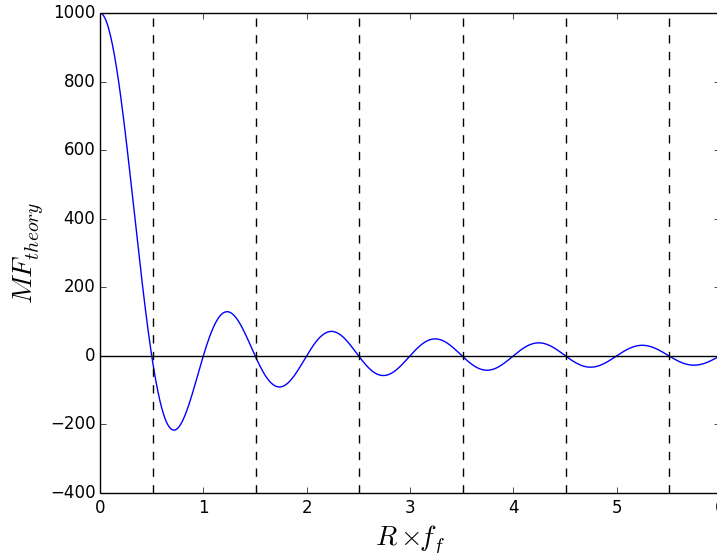


Figure 22: This figure shows the Bessel function for our Moon data comparing MF_{theory} to $R \times f_f$ in order to calculate the radius. Values used for $R \times f_f$ are when the function above crosses the x-axis.

4.2.3 Measuring the Diameter

When measuring the diameter, because of my specific method of envelope detection, it was not necessary to find ϕ from Equation 20 on page 17. Instead, I ended up performing a nonlinear least square fitting to fit a polynomial to my data and effectively return the Bessel function associated to the data, the fringe frequency, and the approximated radius.

Object Name:	HA	δ	f_f (Hz)	Diameter	Actual Diameter
Sun	-0.8865	0.7282	167.9125	30.9148'	31.6' – 32.7'
Moon	1.4045	0.0000	58.9277	29.7526'	29.3' – 34.1'

Table 1: Hour angle, declination, fringe frequency, and source radius calculated from Sun and Moon data.

As shown in the table above, the fringe frequencies and diameters were close to what was to be expected. The Sun, however, was about 2.1684% off from the average low. To approximate the right diameter for the Sun, I used the second $R \times f_f$ value returned from the Bessel function shown in Figure 19 on page 19. For the Moon, I used the first value shown on the Bessel function.

4.3 Point Source Data

4.3.1 The Orion Nebula

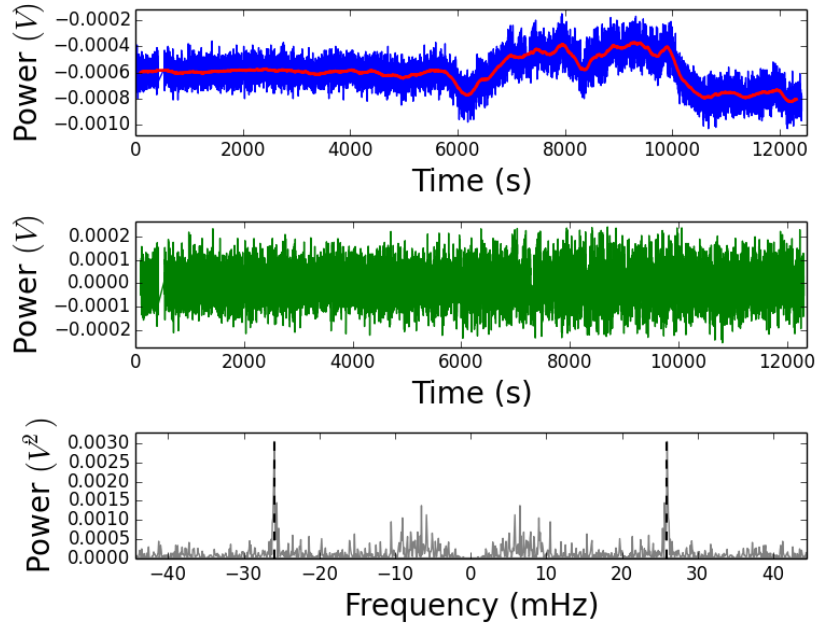


Figure 23: Treated data with power spectrum from the Orion Nebula. Power leakage occurred at about 25.96mHz.

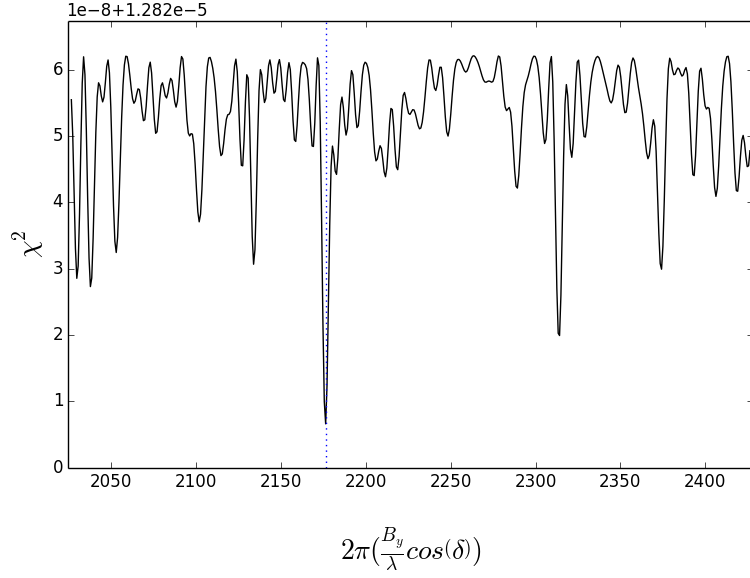


Figure 24: Calculation of $\frac{B_y}{\lambda} \cos(\delta)$ for the Orion using the “Least Square Fitting” method. Least squares done on data ranged from about $x = 2300s$ to $5000s$. The dotted-line shows what value made the best approximation to the baseline from the data.

4.3.2 M17 Nebula

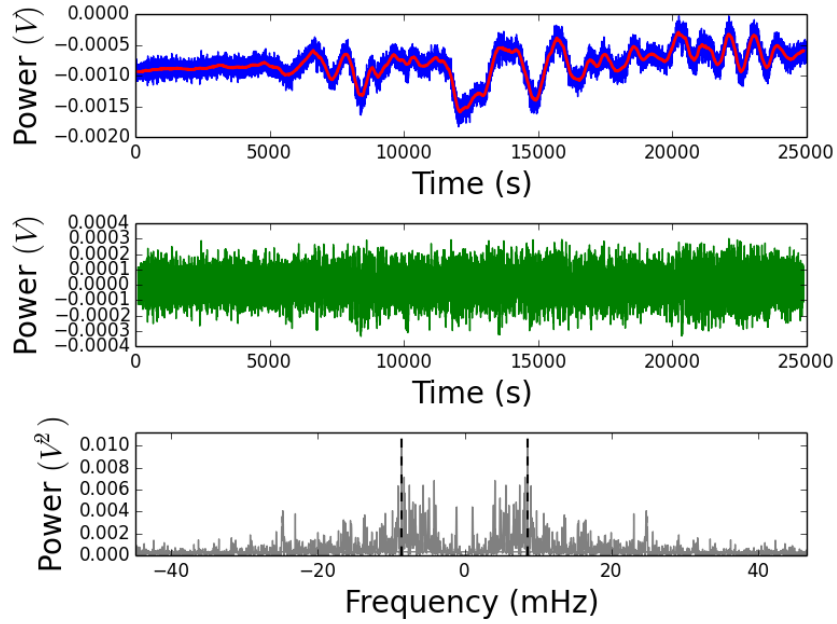


Figure 25: Treated data with power spectrum from the M17 Nebula. Power leakage occurred at about 8.58mHz.

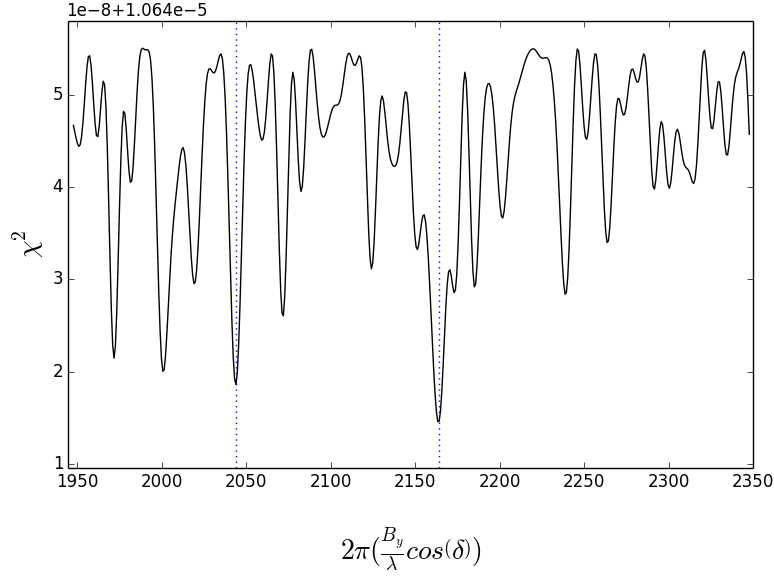


Figure 26: Calculation of $\frac{B_y}{\lambda} \cos(\delta)$ for the M17 using the “Least Square Fitting” method. Least squares done on data ranged from about $x = 3606s$ to $5604s$. The two dotted-lines acrossing shown in blue appeared to be fairly close to each other. However, since the lowest χ^2 value shown on the graph still returned a closer approximation to the baseline, it was used for my calculations.

4.3.3 Virgo A

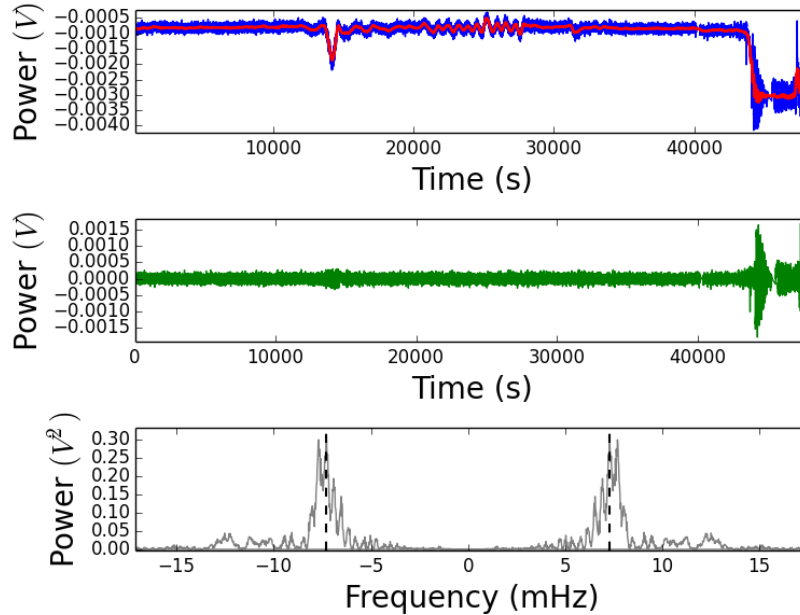


Figure 27: Treated data with power spectrum from Virgo A.

Object Name:	r.a.(2000)	dec(2000)	S_{J_y}
Orion Nebula	$05^h35^m17.3s$	$-05^\circ23'28''$	~ 340
3C274 (Virgo A)	$12^h30^m49.423s$	$+12^\circ23'28.04''$	~ 34
M17	18^h20^m26s	$-16^\circ10.6'$	~ 500
Sun	varies	varies	
Moon	varies	varies	

Table 2: 3C144 Data Table for point sources and the Sun and Moon provided on the UC Berkeley astronomy webpage AstroBaki.

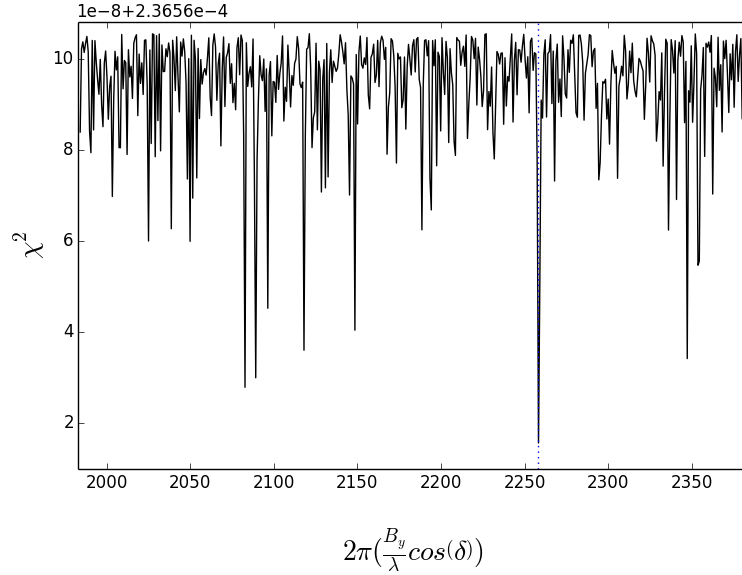


Figure 28: Calculation of $\frac{B_y}{\lambda} \cos(\delta)$ for Virgo A using the “Least Square Fitting” method. Least squares done on data ranged from about $x = 17123s$ to $22130s$. Power leakage occurred at about $7.28mHz$.

4.3.4 Measuring the Baseline

To measure an accurate value for the baseline, this can be done determined by fitting interferometric data for a point source using Equation 13 on page 13. This equation has three unknown quantities; the time delay due to the cable, τ_c , the baseline B_y , and the declination of the target δ . To determine the baseline, my partners and I referred to the documented declination of of our point sources seen in Table 2. Table 3 shown below has the derived coefficients for Equation 13 on page 13 and baseline calculated from using least square fitting on our point source data.

Object Name	A	B	χ^2	Baseline(B_y)	ΔB_y	% Error
Orion	-2.1816e-06	2.4396e-06	2176.2744	9.7750	0.2250	2.2503
M17	2.0591e-06	-2.2544e-06	2164.1595	10.0765	-0.0765	0.7651
Virgo A	3.7002e-07	4.3418e-07	2258.3223	10.3395	-0.3395	3.3948

Table 3: This table shows the resulting coefficients A and B from Equation 13 on page 13 using least square fitting along with the accuracy of fit or square of the residual χ^2 . From these results, the baseline information and relative error was derived.

The average baseline was approximately 10.0637-m, a value within 0.6365% of the initial 10-m guess. Errors in data result may have resulted from bad data measurements, possible (but not likely) flaws in coding, external signal noise from the location of the telescopes, and/or improper selection of data for the least square fitting.

5 Conclusion

Interferometry is an extremely useful tool for making more precise measurements than from using single radio dishes. By synchronizing two radio telescopes separated by a baseline, this effectively increases the resolution of data measurements. Since there is a baseline separation when using the interferometer, there is a distinct geometric delay between the two dishes. This geometric delay is what produces the “fringe patterns” we observed. Fringe patterns can be used to determine either the baseline of the interferometers or declination of an astronomical point source being tracked.

When measuring the Sun or the Moon with the interferometer, the size of the source is relatively larger than the interferometer we used. Because of this, this gives rise to a modulating function. From the modulating function, the angular radius can then be determined from the function’s shape. Although our data could have been better, results were still fairly close to what we expected. Ultimately, this lab has taught how to properly analyze signals and data in Python, use an interferometer to make measurements, and approximate the baseline from point sources and angular diameter for the Sun and Moon.

6 Acknowledgements and References

1. Lab Partners: Aaron Tran, Leonardo Cassara, and Ryan Gao.
2. Additional help with analysis, coding, and the report was also provided by Baylee Bordwell and Eduardo Herrera.
3. Undergraduate radio lab. (2014, April 8). Retrieved from https://casper.berkeley.edu/astrobaki/index.php/Undergraduate_Radio_Lab.
4. Additional pictures were taken from Wikipedia, the free online encyclopedia.